Assignment #5 : Recurrence + HashTables

$\boxed{\#1}$ $T(n) = 5T(\frac{n}{7}) + 3n^2$

- $T(\frac{n}{7}) = 5T(\frac{n}{7^2}) + 3(\frac{n}{7})^2$

$T(n) = 5[5T(\frac{n}{7^2}) + 3(\frac{n}{7})^2] + 3n^2 = 25T(\frac{n}{7^2}) + 15(\frac{n}{7})^2 + 3n^2$

- $T(\frac{n}{7^2}) = 5T(\frac{n}{7^3}) + 3(\frac{n}{7^2})^2$

$T(n) = 5[5[5T(\frac{n}{7^3}) + 3(\frac{n}{7^2})^2] + 3(\frac{n}{7})^2] + 3n^2$

$= 125T(\frac{n}{7^3}) + 75(\frac{n}{7^2})^2 + 15(\frac{n}{7})^2 + 3n^2$

$5^1 \times 3$

$5^2 \times 3$

$\boxed{T(n) = 5^j T(\frac{n}{7^j}) + \sum_{i=1}^{j} \left(\frac{5^{i-1}}{7^{2(i-1)}}\right) \cdot 3n^2}$

$\hookrightarrow \left(\sum_{i=1}^{j} \frac{(5)^{i-1}}{7^{2(i-1)}}\right) 3n^2$

Master Method

$a = 5$, $b = 7$, $f(n) = 3n^2$, $d = 2$

$a \ ? \ b^d \rightarrow 5 < 7^2 \Rightarrow \Theta(n^2)$

1)

2)

$a \geq 1, \; b > 1 \quad d \geq 0$

$T(n) = aT(\frac{n}{b}) + f(n)$

#2 Master Theorem

a) $T(n) = 5T(\frac{n}{9}) + n^7$

- $f(n) = n^7, \; d = 7, \; a = 5, \; b = 9$

$a \; ? \; b^d = 5 < 9^7 \Rightarrow \boxed{T(n) = \Theta(n^7)}$

b) $T(n) = 3T(\frac{n}{3}) + 7n$

- $f(n) = 7n, \; d = 1, \; a = 3, \; b = 3$

$a \; ? \; b^d = 3 = 3^1 \Rightarrow \boxed{T(n) = \Theta(n \log n)}$

c) $T(n) = 15T(\frac{n}{4}) + 100$

- $f(n) = 100(n^0), \; d = 0, \; a = 15, \; b = 4$

$a \; ? \; b^d = 15 > 4^0 \Rightarrow \boxed{T(n) = \Theta(n^{\log_4(15)})}$

d) $T(n) = 9T(\frac{n}{3}) + n^2$

- $f(n) = n^2, \; d = 2, \; a = 9, \; b = 3$

$a \; ? \; b^d = 9 = 3^2 \Rightarrow \boxed{T(n) = \Theta(n^2 \log n)}$

e) $T(n) = 6T(\frac{n}{5}) + n^3$

- $f(n) = n^3, \; d = 3, \; a = 6, \; b = 5$

$a \; ? \; b^d = 6 < 5^3 \Rightarrow \boxed{T(n) = \Theta(n^3)}$

Because the functions are in the form of T(n) = aT(n/b) + f(n), and all components meet the criteria (a >= 1 and b > 1), we can you the Master Theorem to solve.

3) Radix Sort - (i know this is incorrect , i switched which side is null, did not have time to fix, I'm sorry!)

BOY, RIS, GUA, ATL, SUC, CES, QUI, PUT, ANT, VAU, HUI, VAL, SAN, SAN, CAQ, VIC, CAL, LA_GUA, TOL, META, CAU, BOL, AMA, MAG, CHO, NSA, GUA, CAS, COR

Iteration 1

NULL	

OTHER	
A	GUA, LA_GUA, META, AMA, NSA, GUA,
B	
C	SUC, VIC,
D	
E	
F	
G	MAG,
H	
I	QUI, HUI,
J	
K	
L	ATL, VAL, CAL, TOL, BOL,
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N	SAN, SAN,
O	CHO,
P	
Q	CAQ,
R	COR,
S	RIS, CES, CAS,
T	PUT, ANT,
U	VAU, CAU,
V	
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Y	BOY,
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OTHER	

GUA, LA_GUA, META, AMA, NSA, GUA, SUC, VIC, MAG, QUI, HUI, ATL, VAL, CAL, TOL, BOL, SAN, SAN, CHO, CAQ, COR, RIS, CES, CAS, PUT, ANT, VAU, CAU, BOY

Iteration 2

NULL	
OTHER	
A	MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU,

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E	CES,
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H	CHO,
I	VIC, RIS,
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M	AMA,
N	ANT,
O	TOL, BOL, COR, BOY,
P	
Q	
R	
S	NSA,
T	META, ATL,
U	GUA, LA_GUA, GUA, SUC, QUI, HUI, PUT,
V	
W	
X	
Y	
Z	
OTHER	

MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU, CES, CHO, VIC, RIS, AMA, ANT, TOL, BOL, COR, BOY, NSA, META, ATL, GUA, LA_GUA, GUA, SUC, QUI, HUI, PUT

Iteration 3

NULL	
A	AMA, ANT, ATL,
B	BOL, BOY,
C	CAL, CAQ, CAS, CAU, CES, CHO, COR,
D	

E	META,
F	
G	GUA, LA_GUA, GUA,
H	HUI,
I	
J	
K	
L	
M	MAG,
N	NSA,
O	
P	PUT,
Q	QUI,
R	RIS,
S	SAN, SAN, SUC,
T	TOL,
U	
V	VAL, VAU, VIC,
W	
X	
Y	
Z	
OTHER	

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, META, GUA, LA_GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC

Iteration 4

NULL	AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC
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OTHER	LA_GUA

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META, LA_GUA

Iteration 5

NULL	AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN,w SAN, SUC, TOL, VAL, VAU, VIC, META
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OTHER	

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META, LA_GUA

NULL	AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META
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OTHER	

AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META, LA_GUA

NULL	AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META, LA_GUA
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Z	
OTHER	

FINAL LIST : AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI, MAG, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC, META, LA_GUA

4)

#4 Double Hashing

[62, 62, 69, 70, 4, 103, 85, 27, 86, 40, 96, 77, 23]

Key : 62

$(|62-9|) \times (62+13) = 53 \times 75 = 3975 \div 15 = 265 + 62 = 327 \% 7$

$= 5$ [homeslot]

0	
1	
2	
3	
4	
5	62
6	

Key 62 → can't have keys that are the same

Key 69

$(|69-9|) \times (69+13) = 60 \times 82 = 4920 \div 15 = 328 + 69 = 397 \% 7$

Collision #1 → $= 5$ [homeslot]

328510

$h_1(69) = (\text{hash}(69) + \text{power}(69)) \% 7$

$= (5 + 328510) \% 7 = 328515 \% 7 = 5 \rightarrow$ Collision #2

$h_2(69) = (\text{hash}(69) + 2\,\text{power}(69)) \% 7$

$5 + 2(328510) \% 7 = (5 + 657020) \% 7 = 657025 \% 7$

$= 5 \rightarrow$ Collision #3

Resize → $7 \times 2 = 14$ → next prime # = 17, M=17

0	
1	86
2	77
3	85
4	
5	62
6	69
7	27
8	
9	4
10	
11	23
12	
13	40
14	96
15	103
16	70

Key 69 (cont)

$(|69-9|) \times (69+13) = 60 \times 82 = 4920 \div 15 = 328 + 69 = 397 \% 17 = 6$ [see final table]

Key 70

$(|70-9|) \times (70+13) = 61 \times 83 = 5063 \div 15 = 337 + 70 = 407 \% 17$

$= 16 \rightarrow$ [homeslot]

Key 4

$(|4-9|) \times (4+13) = 5 \times 17 = 85 \div 15 = 5 + 4 = 9 \% 17 = 9 \rightarrow$ (homeslot)

Key 103

$(|103-9| \times (103+13)) = 92 \times 116 = 10672 \div 15 = 711 + 103 = 814 \% 17$

$= 15 \rightarrow$ (homeslot)

Key 85

$(|85-9|) \times (85+13) = 7448 \div 15 = 496 + 85 = 581 \% 17 = 3 \rightarrow$ [homeslot]

Key 27

$(|27-9|) \times (27+13) = 720 \div 15 = 48 + 27 = 75 \% 17 = 7 \rightarrow$ (homeslot)

Key 86

$(|86-9|) \times (86+13) = 7623 \div 15 = 508 + 86 = 594 \% 17 = 16 \rightarrow$ (homeslot)

↳ Collision #1

$h_1(86) = (\text{hash}(86) + \text{power}(86)) \% 17$

$(16 + 636057) \% 17 = 636073 \% 17 = 1$ [see final table]

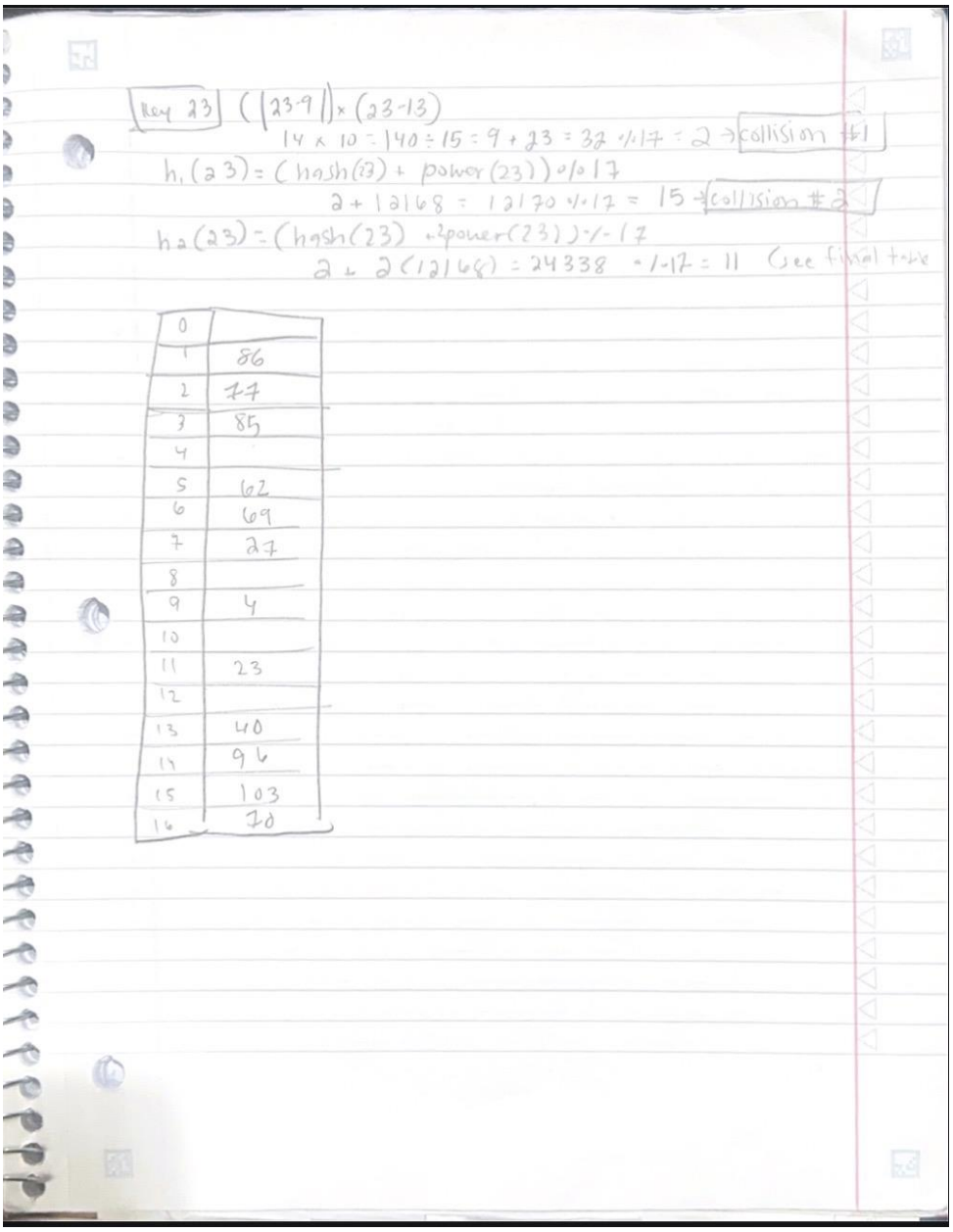
Key 40 $(|40-9|) \times (40+13) = 1643 \div 15 = 109 + 40 = 149 \% 17 = 13$ [homeslot]

Key 96 $(|96-9|) \times (96+13) = 9483 \div 15 = 632 + 96 = 728 \% 17 = 14$ [homeslot]

Key 77 $(|77-9|) \times (77+13) = 402 \div 15 = 26 + 77 = 103 \% 17 = 1 \Rightarrow$ Collision #1

$h_1(77) = (\text{hash}(77) + \text{power}(77)) \% 17$

$1 + 4914 = 4915 \% 17 = 2$ (see final table)



Key 23 $(|23-9|) \times (23-13)$

$14 \times 10 = 140 \div 15 = 9 + 23 = 32 \% 17 = 2 \rightarrow$ collision #1

$h_1(23) = (\text{hash}(23) + \text{power}(23)) \% 17$

$2 + 12168 = 12170 \% 17 = 15 \rightarrow$ collision #2

$h_2(23) = (\text{hash}(23) + 2\text{power}(23)) \% 17$

$2 + 2(12168) = 24338 \% 17 = 11$ (see final table

0	
1	86
2	77
3	85
4	
5	62
6	69
7	27
8	
9	4
10	
11	23
12	
13	40
14	96
15	103
16	70

7) Algorithm Analysis

Hashing: The time complexity would be O((N-1)*M) where N is the length of the inputted array and M is the size of the hash table. For each insertion (N), you could have a collision at all insertions after the first one (N-1), and each collision is O(M) complexity because the worst case would involve having to collide with all buckets. Thus, the time complexity is O((N-1)*M). The space complexity is O(M), where M is the size fo the hash table because size of the table may based on the number of collisions, and to rehash, the M value may change.

Radix Sort:

The time complexity is O(N*M), where N is the length of the longest string and M is the length of the inputted array. In the algorithm, the there are a multitude of for-loops, they either run the length of the input array (which has M number of elements), 256 times, and the number of elements in each bucket (which is N). The outer loop in radiz sort runs N times, and for each iteration, sortBuckets is called. In sortBuckets, the first loop runs 256 times, the second loops runs M times, and the third loop runs 256 + M times. Because 256 is constant, and we are doing asymptotic analysis, we can drop it, so we have O(M) operations in sort buckets being called N times in the overall algorithm. Thus, the time complexity is O(N*M).

The space complexity O(N+M) where N is the length of the longest string and M is the length of the inputted array because we create 256 buckets (unchanging for input so it is constant), and all buckets (together) hold M number of strings . Because the maximum length of the strings (maxString) is dependent on the inputted array (at worst is N), the space complexity is O(N+M)

Palindrome Triplets:

The time complexity is O(N^3 *M), where N is the length of the inputted array and M is the length of the longest string. There are three nested for loops, and the complexity of a loop is the loop itself * its body. Each of the “loops it self” run N times, creating a time complexity of O(N^3). In the for most inner for loop, the isPalindrome is being called, and

The space complexity is O(N * M) where N is the length of the inputted array in the inputted array and M is the length of the longest string because resultList depends on the elements of the inputted array , and it,in the worst case, hold all combinations of the words. The maximum length of the combined string is based off of the length of the longest string (M), and could be as long as 3M. We can ignore the 3 because we are doing asymptotic complexity. So, the space complexity is O(N*M).